

Comparative Analysis of Artificial Intelligence Performance Across Multimodal Tasks

This research presents a comprehensive comparative evaluation of three advanced artificial intelligence models ChatGPT, Gemini, and Copilot tested across multiple performance dimensions. All models examined in this study were used in their freely accessible, non-premium versions to ensure equal testing conditions. The primary objective is to analyze their efficiency, technical proficiency, and interpretative accuracy across three principal skill domains: visual generation, software development, and mathematical reasoning. A Photo Generation Test Results.

Each AI model was given two visual generation prompts to assess its creative capabilities. Performance was evaluated in three categories: response speed, technical skill, and comprehension accuracy.

Prompt for the First Image:

A man is seated on a wooden chair in a dimly lit vintage room, wearing an oversized jacket in a skin tone hue. Sunlight filters through the window blinds, casting dramatic striped shadows across his face and the surrounding wall. The atmosphere evokes a nostalgic and contemplative mood, enriched with a soft golden hue and warm tonal palette. The facial features should remain consistent, and the image should maintain a 9:16 aspect ratio.

Photo Generation – Test 1

Model	Speed (s)	Skill	Comprehension	Overall Score
Gemini	10	8	10	9.3 / 10
ChatGPT	15	6	10	8.0 / 10
Copilot	140	8	10	6.6 / 10

Gemini demonstrated superior speed and overall quality, while ChatGPT showed strong interpretative understanding. Copilot produced acceptable results but with significantly longer response times.

Prompt for the Second Image:

Transform the original image into a black-and-white portrait captured from a three-quarter angle, with the subject's body slightly turned while the face remains directed to the left. The expression should convey calm contemplation or deep emotion. Employ a soft, single source studio light from one side a cinematic side-lighting technique to create a high contrast chiaroscuro effect, illuminating one half of the face while the other remains dramatically shadowed.

Photo Generation – Test 2

Model	Speed (s)	Skill	Comprehension	Overall Score
Gemini	8	8	6	8.0 / 10
ChatGPT	10	4	8	7.0 / 10
Copilot	135	7	6	5.0 / 10

In the second test, Gemini again outperformed its peers with superior speed and balanced comprehension. ChatGPT displayed logical understanding but lower visual fidelity, while Copilot lagged behind in speed.

Overall Averages (Photo Generation)

Model	Avg. Speed (s)	Avg. Skill	Avg. Comprehension	Overall Avg. Score
Gemini	9	8	8	8.65 / 10
ChatGPT	12.5	5	9	7.5 / 10
Copilot	137.5	7.5	8	5.8 / 10



Photo1 Chat Gpt



Photo1 Gemini

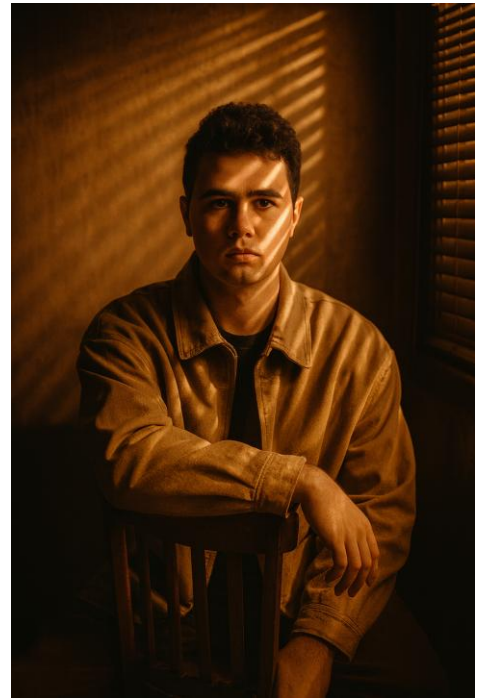


Photo1 Copilot



Photo2 Chat Gpt

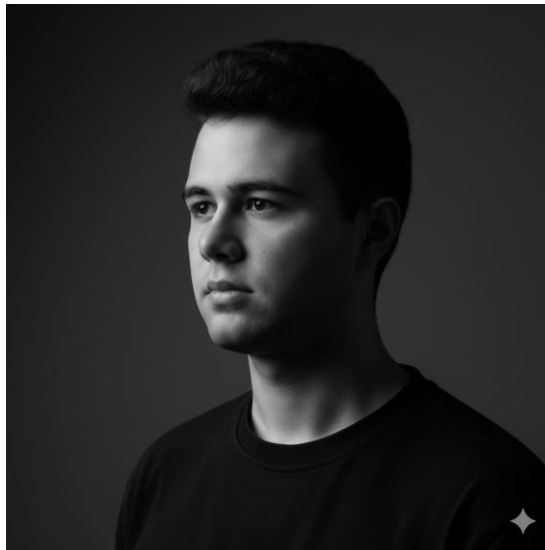


Photo2 Gemini



Photo2 Copilot

B. Software Development and Setup Test

Each model was tasked with building a C# To Do List application using Visual Studio Community. The task required the AI to guide a non-technical user through the setup and coding process.

Gemini achieved the fastest completion time (8 seconds) and integrated interactive checkboxes, enhancing user engagement. ChatGPT produced a more logically complete structure but lacked visual interactivity. Copilot created a functional but basic implementation.

ChatGpt: Developed a functional “To Do List” application that allows users to add and delete tasks. The application operates effectively, though it remains minimal in terms of visual design and interactive features.

Gemini: Integrated clickable checkboxes, enhancing user engagement and interactivity. However, the functional depth of the application was relatively limited.

Copilot: Did not include checkboxes or interactive elements; it was restricted to generating a basic, static task list.

Model	Speed (s)	Skill	Comprehension	Overall Score
Gemini	8	5	7	7.3 / 10
ChatGPT	37	6	5	5.7 / 10
Copilot	12	4	4	5.7 / 10

C. Mathematical Reasoning and Logical Analysis

Each model was presented with a real-world percentage and tax calculation problem. All three AIs successfully reached the correct final answer (176 TL (Turkish Lira)).

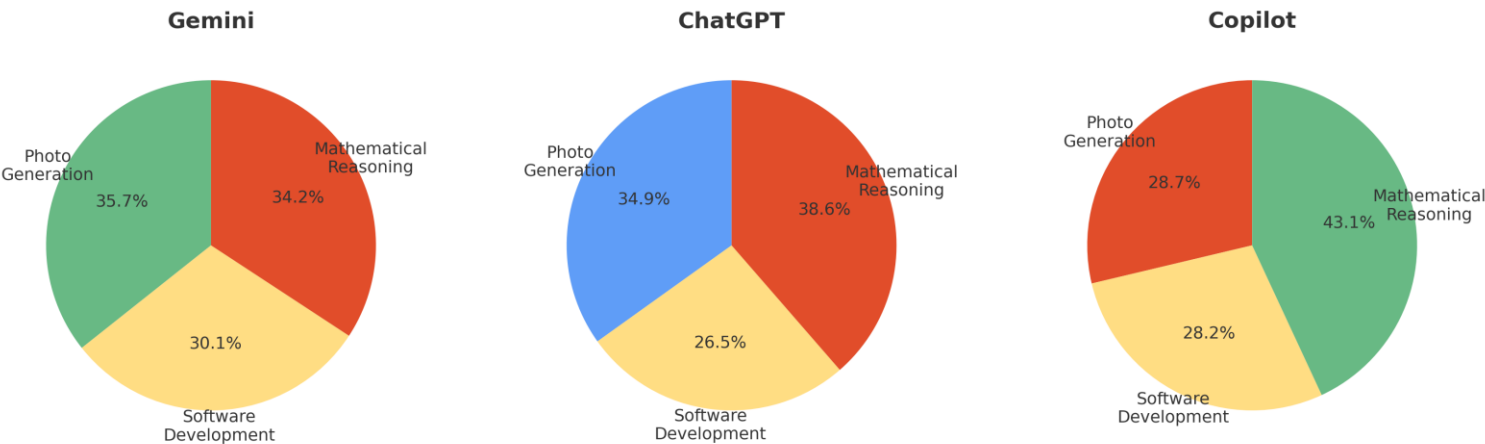
Model	Speed (s)	Skill	Comprehension	Overall Score
Copilot	5	10	10	8.7 / 10
ChatGPT	7	9	9	8.3 / 10
Gemini	7	9	9	8.3 / 10

Overall Performance Summary

Model	Photo Generation	Software Development	Mathematical Reasoning	Overall Average
Gemini	8.65	7.3	8.3	8.1 / 10
ChatGPT	7.5	5.7	8.3	7.2 / 10
Copilot	5.8	5.7	8.7	6.7 / 10

Visual Representation of AI Competence Levels

AI Skill Comparison
(Photo Generation - Software Development - Mathematical Reasoning)



Conclusion and Analytical Report

The comparative study reveals that all three AI systems demonstrate high analytical and computational accuracy, yet they vary in their execution speed, depth of reasoning, and expressive sophistication. Gemini consistently excels in visual and interactive responsiveness, while ChatGPT offers a balanced synthesis of logic and communication clarity. Copilot, although technically robust in mathematical processing, lags behind in creative and interactive tasks.

In conclusion, Gemini stands out as the most versatile model across multimodal tasks, ChatGPT excels in structured reasoning and explanation, and Copilot performs best in rapid, computation-driven operations. Creative and interactive tasks